
Monocentric City Model I

LSE GY457 - AT 2025

Topic 4

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Course overview

- Block 1

- Intro to Urban and Regional Economics and course overview
- Topic 1: Regional and urban concentration forces
- Topic 2: The empirics of agglomeration
- Topic 3: The costs and benefits of agglomeration

- Block 2

- **Topic 4&5: Monocentric city model**
- Topic 6: Firm location choice

- Block 3

- Topic 7: Neighborhoods and segregation
- Topic 8: The vertical dimension of cities
- Topic 9: Gentrification and suburbanization
- Topic 10: Measuring cities with new technologies

Recap from last time

- Cities and the “fundamental trade-off of urban economics”
 - Higher density associated with a range of costs and benefits
- Equilibrium city size
 - Cities tend to be too big given the local costs and benefits
 - ...but potentially too small given the global externalities
- Compensating differentials
 - Productive cities have high wages and high rents
 - More amenable (livable) cities have higher rents relative to wages

Today: Monocentric city model I

- Introduction to monocentric city model
- Household bid-rent model
 - Simple Alonso (1964) model
 - Adding the construction sector: Mills (1967) and Muth (1969)
- Empirical patterns
 - (Urban) rent gradient

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History of thought: Ricardo (1817)

- Why paying higher prices for better locations?
- Higher prices must offset locational advantages
- Ricardo (1817): founder of “compensating differentials”
 - Bid higher rents for more fertile land to equalize profits
 - Spatial competition ensures zero profits
 - Land rent compensates for locational advantages

History of thought: Von Thünen (1826)

- vs. Ricardo: replace land fertility with land *access*
- Lower transportation costs of shipping agricultural products to the central (urban) market increase profits, hence accessibility must be compensated with higher rents
 - Framework already features key points of modern “bid-rent” models
 - Assumption: Higher land rent offsets lower transportation costs
 - Prediction: Prices decline with distance from the center

History of thought: Alonso (1964)

- Generalizes Von Thünen concept to the urban context
- That is: Replace goods transportation costs with worker commuting costs
 - ...Assuming that residents consume land directly
- In spatial equilibrium, utility must be equalized across locations
 - Commuting cost differences in urban area are mirrored by price differences
 - Prediction: Prices decline and dwelling sizes (plots of land) increase with distance from the center

History of thought: Mills (1967) and Muth (1969)

- Build on Alonso (1964), adding competitive construction sector
- Land and structural capital are intermediate inputs in the production of housing
- In spatial equilibrium, utility must be equalized across locations
 - Competitive construction sector implies zero profits
 - Prediction: Prices, dwelling sizes, *and structural density* (governing building heights) decline with distance from the city center

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AMM model: Intuition

- Rents adjust to make people indifferent between locations
- Standard model: Locations differ because of commuting costs; city center offers a fixed wage
- In addition to explaining prices, can explain density if we allow density to vary (e.g., skyscraper model)

Building blocks

- Simple version of the Alonso model
 - Topic 4 (today)
- + Adding developers (Muth and Mills models)
 - Topic 4 (today)
- Income segregation, open city, comparative statics
 - Topic 5
- Firm location choice (from Fujita and Ogawa)
 - Topic 6
- Urban model in general equilibrium (*hint*)
 - Topic 6

Alonso bid-rent theory: Rent definition

- Definition: “Pure economic rent” is the price paid per year (month) for the services of a factor of production that is fixed in supply (McDonald and McMillen, 2007)
 - E.g. for urban land, agricultural land

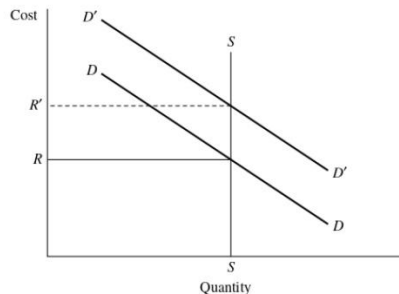


Figure 6.1 Determination of pure economic rent

- Market value (V) equal to the stream of rents that the asset will earn:
$$V = \frac{R_1}{1+r} + \frac{R_2}{(1+r)^2} + \frac{R_3}{(1+r)^3} + \dots$$
- If rent is expected to be the same each year:
$$V = \sum_{i=1}^n \frac{R}{(1+r)^i}$$
- If asset life is infinite: $V = R/r$
 - Asset value proportional to its rent

Alonso bid-rent theory: Assumptions

- A1: (Costly) access to the (exogenous) Central Business District (CBD) is the only locational advantage
- A2: Plain featureless ground
- A3: Identical households and firms
- A4: No government
- A5: No developers (households and firms occupy land)
- A6: Population fixed (no migration)

Simplifications needed to derive testable hypotheses and evaluate counterfactual scenarios

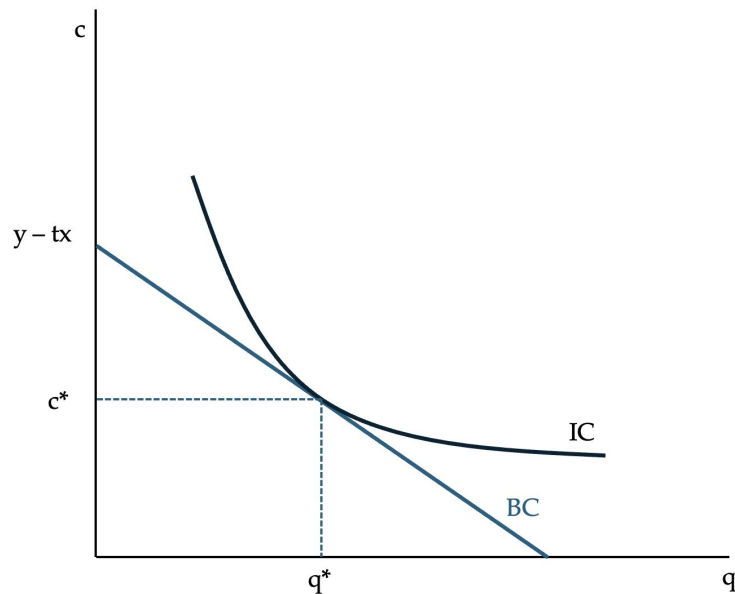
Alonso bid-rent theory: Framework

- Households maximize utility U
- Households consume a basket of goods (c) and land (q)
 - $U = U(c, q)$
 - $U' > 0$; $U'' < 0$ (utility function is concave in both its arguments)
- Households must commute to the CBD to work (costly!)
 - Face commuting costs (at rate t over distance x from CBD)
 - They supply one unit of work inelastically and get fixed wage y
- Household budget constraint
 - $y - tx = p_q q + p_c c$
 - Fix $p_c = 1$; then: $y - tx = p_q q + c$ (relative prices is all that matters)
 - They spend budget on commuting and on consumption of c and q

Alonso bid-rent theory: Equilibrium

- Spatial equilibrium condition: Households must be indifferent between locations
 - No household has an incentive to move to a different location
 - In equilibrium, utility in each location is fixed (equal to $\bar{U}$)
- Land rent adjusts to compensate for transport cost saving
 - Intuitively, if $x \uparrow p_q \downarrow$ and vice versa
 - Formally: Solve the budget equation for p_q : $p_q = (y - tx - c)/q$
 - Compute partial derivative of p_q wrt x : $\partial p_q / \partial x = -t/q < 0$
 - Rents p_q decrease with distance from the CBD x
 - What happens to land consumption as we move away from CBD?

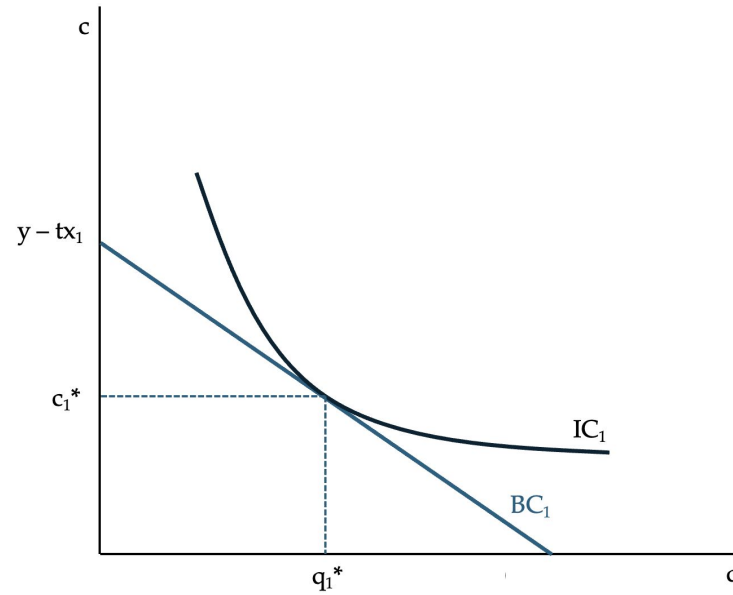
Alonso bid-rent theory: Household choice problem



- Solve budget constraint (BC) for c : $c = y - tx - p_q q$
- Slope of budget constraint equal to $-p_q(x)$
- Intercept of budget constraint ($c | q=0$) depends on commuting costs (tx)

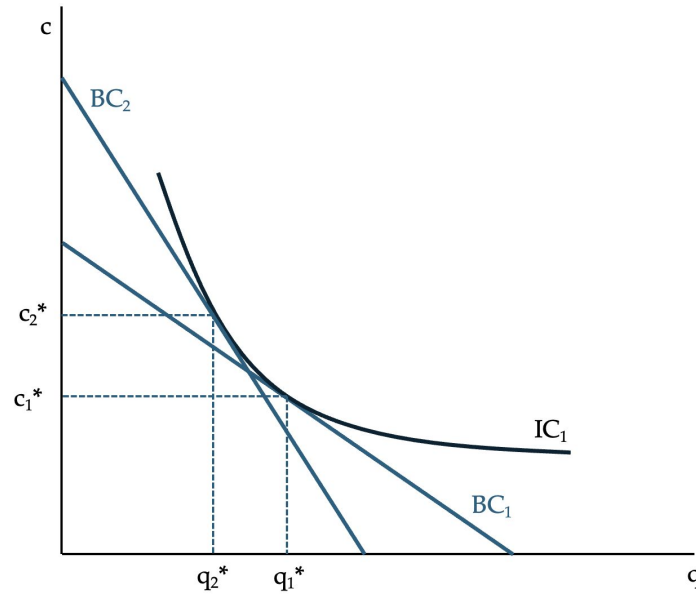
>>> Spatial equilibrium implies that the shape of the budget constraint depends on distance to CBD (x)!

Alonso bid-rent theory: Household choice problem



- Let's compare two locations, at distance x_1 and x_2 from CBD
- At distance x_1 , household has $y - tx_1$ to spend on (c, q)
- In location 1, equilibrium quantities are c_1^* , q_1^* (where indifference curve IC_1 intersects BC_1)

Alonso bid-rent theory: Household choice problem

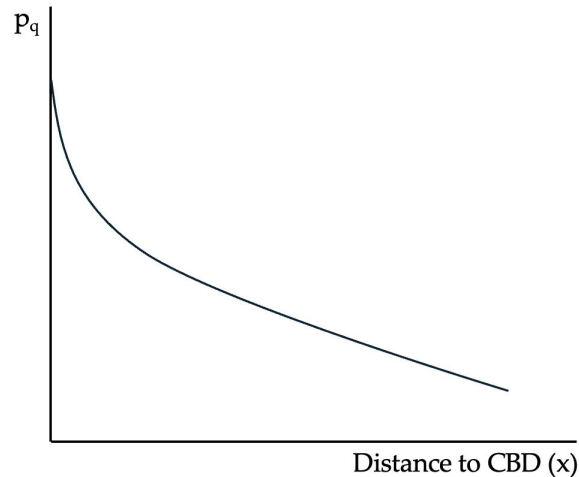


- Consider now what happens if the household were in location 2 (at distance x_2 from CBD)
- The budget constraint for location 2 is denoted BC_2
- Which of the two locations (1 vs. 2) is more central and which one is peripheral? Which one has higher rent?

Alonso bid-rent theory: Predictions

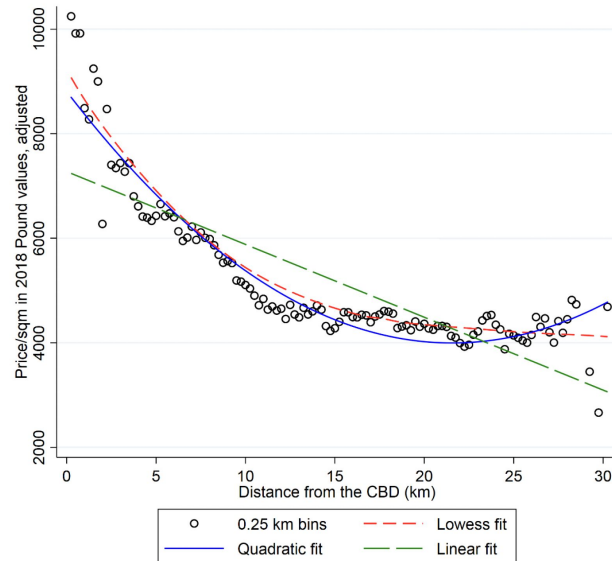
- As distance from the CBD increases ($x \uparrow$):
 - Rent decreases $p_q \downarrow$: $\partial p_q / \partial x = -t/q < 0$
 - What happens to land consumption (q)?
 - What happens to population density $D=1/q$? (Also: why $D=1/q$)?
- Further predictions: Rents (p_q)
 - ... decrease in commuting costs: $\partial p_q / \partial t = -x/q < 0$
 - ... increase in income: $\partial p_q / \partial y = 1/q > 0$

Alonso bid-rent theory: Rent gradient



- Bid rent function: how much household is willing to pay for land at different distances from CBD
- Bid-rent negative function of distance to the CBD
 - Curvature of the function depends on degree of substitution between c, q (the higher the substitutability, the more curved the function)

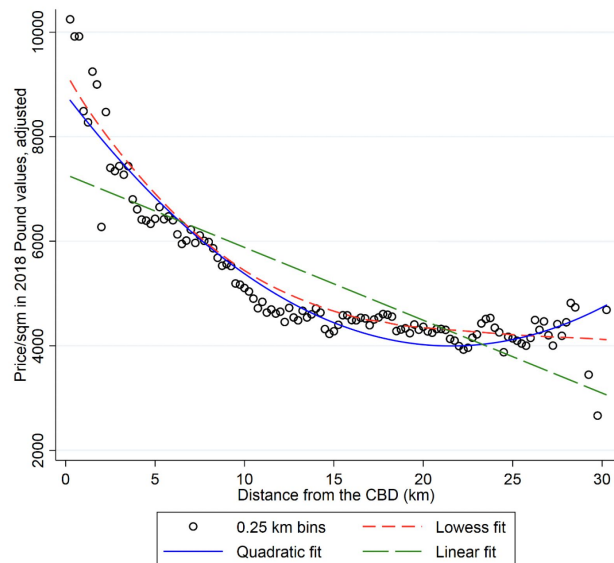
House prices per sqm in London



Based on hedonic analysis of 1995-2008 transaction data.
Adjusted to 2018 prices using the Land Registry London house price index
(c) Ahlfeldt, 2018

- Average house price at given (bin) distance from CBD
- Pattern in line with theory!

House prices per sqm in London



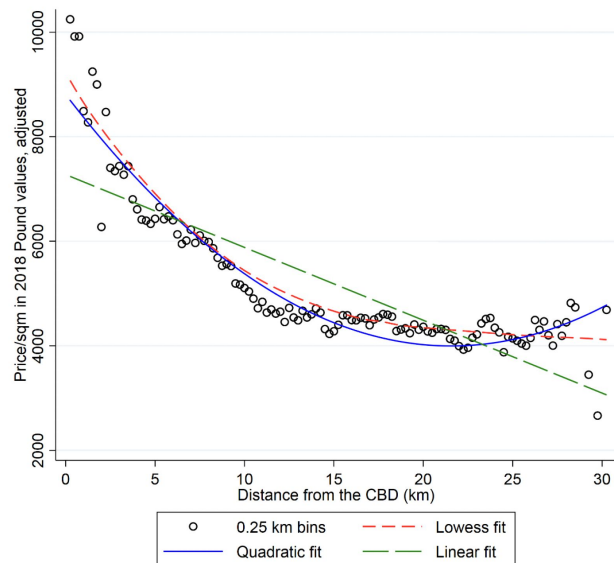
Based on hedonic analysis of 1995-2008 transaction data.
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(c) Ahlfeldt, 2018

- How is this graph built?

- 1 *Hedonic method*

- Regress house prices on a set of variables reflecting property characteristics (e.g., structure age, heating system, presence of garage, floor level, etc) *but not distance to the CBD*
 - Recover the fitted values from this regression

House prices per sqm in London



Based on hedonic analysis of 1995-2008 transaction data.
Adjusted to 2018 prices using the Land Registry London house price index
(c) Ahlfeldt, 2018

- How is this graph built?
 - 2 *Collapse info at bin level*
 - Collapse the data (i.e. take the mean) into “bins” (check *binsreg/binscatter* commands)
 - Size of one bin: 250 meters
 - Side note: binned scatter plot useful to visualize correlation between two variables when using large amount of data

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Adding construction sector (supply of housing)

- Alonso (1964): households receive utility directly from land consumption (and other goods)
- vs. Mills (1967) and Muth (1969) add construction sector
 - Even if they worked independently, extension known as “Muth-Mills” model
 - Households get utility from housing (produced by developers using a combination of land and capital) and other goods
 - Incorporate building heights, lot sizes, and house prices to the analysis

Muth-Mills model: Housing production

- Profit maximizing developers produce housing H using a combination of capital (K) and land (L)
- $H = H(K, L)$; $H' > 0$, $H'' < 0$ (concave production function)
- Developers rent housing to households at price given by household bid rent p_q (as before)
- Profit function: $\pi = p_q H(K, L) - p_K K - rL$
 - Price of capital normalized to 1 (i.e., $p_K = 1$); r is unit price of land
- Capital to land ratio: $S = K/L$ (structural density)
 - Governs building heights, but not the same (e.g., S high also when developers invest in high-quality materials or complex designs)

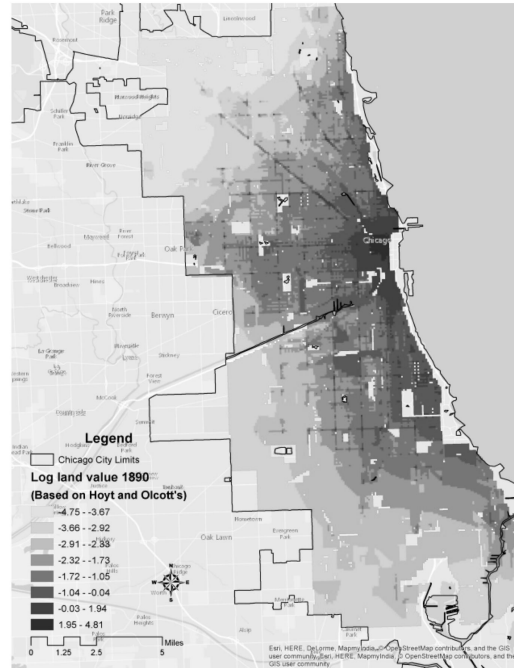
Muth-Mills model: Developer profit-maximization problem

- Perfect competition among developers implies zero profits in equilibrium ($\pi=0$):
- $\pi = p_q H(K,L) - K - rL = 0$
- So that (solving for r) the bid-rent function is:
$$r = (p_q H(K,L) - K)/L$$
- Rewrite the profit function in terms of housing density ($h=H/L$) and structural density ($S=K/L$): $\pi = p_q h(S) - S - r$
 - Find level of S that maximizes profits and set them to 0
 - $\partial\pi/\partial S = p_q h'(S) - 1 = 0 \gggg p_q h'(S) = 1$
 - $p_q \uparrow S \uparrow$ (Formally: $\partial p_q / \partial S = -h''(S) / [h'(S)]^2 > 0$)

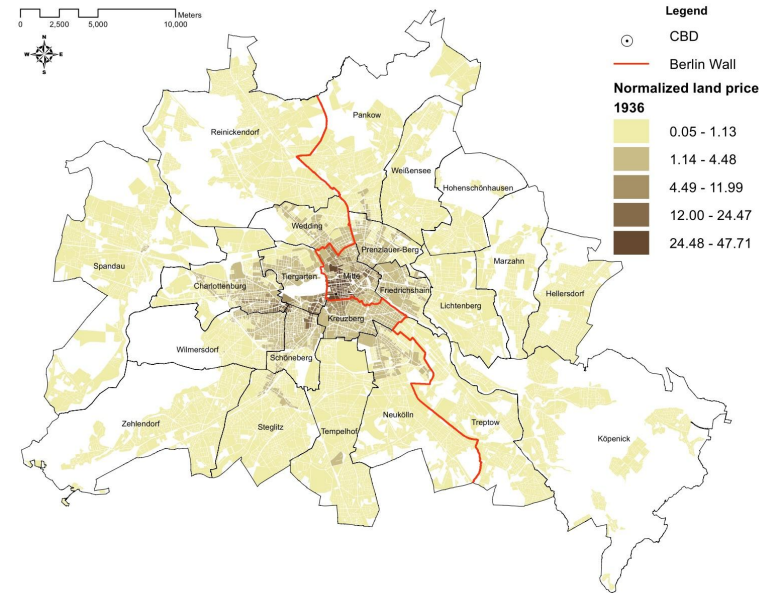
Muth-Mills model: Predictions

- As housing rent increases ($p_q \uparrow$):
 - Land rent increases $r \uparrow$: Developers bid more aggressively for land
 - Structural density ($S = K/L$) increases: as $p_q \uparrow$, $r \uparrow \gg L \downarrow K \uparrow \gg S \uparrow$
 - Housing density increases (H/L) $\uparrow$: Marginal benefit of adding housing units increases $\gg H/L \uparrow$ to maximize units per land area
- As distance from the CBD increases ($x \uparrow$):
 - What happens to structural density, housing density, and land rent?

Land prices in Chicago and Berlin

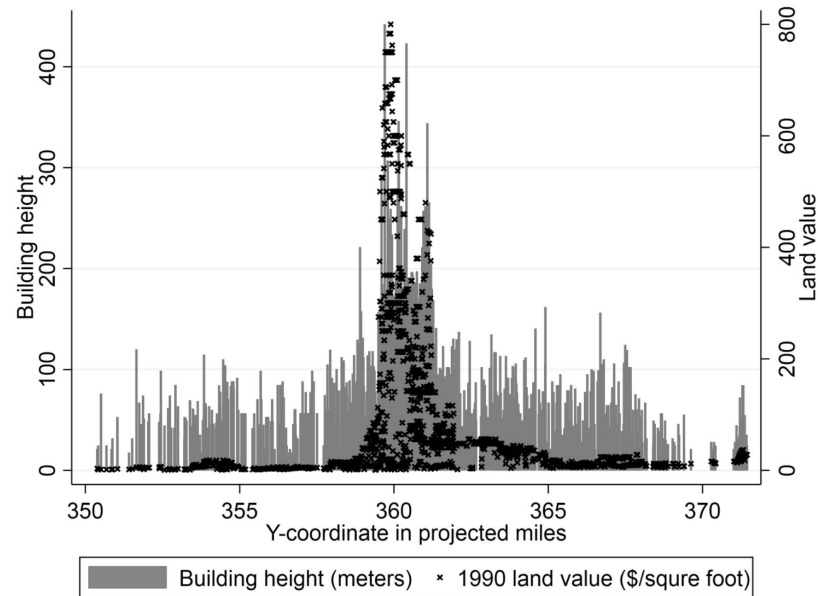


Source: Ahlfeldt and McMillen (2018)



Source: Ahlfeldt et al. (2015)

Building height and land value in Chicago (Ahlfeldt and McMillen, 2018)



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Population density

- In Alonso model: $D(x) = 1/q(x)$
 - $N(x)$ = population in x ; $L(x)$ = land in x
 - $q(x)$ = land per capita = $L(x)/N(x)$
 - $D(x)$ = population density = $N(x)/L(x) \ggg$ i.e. $D(x)=1/q(x)$
- In Muth-Mills model: $D(x) = h(x)/q(x)$
 - $N(x)$ = population in x ; $H(x)$ = housing in x
 - $q(x)$ = space (housing) per capita = $H(x)/N(x)$
 - $h(x)$ = housing density = $H(x)/L(x)$
 - $D(x)$ = population density = $N(x)/L(x) = h(x)/q(x)$
 - To see this: $D(x) = N(x)/L(x) = [N(x)H(x)]/[L(x)H(x)] =$
 $[H(x)/L(x)]*[N(x)/H(x)] = [H(x)/L(x)]*[N(x)/H(x)]^{-1} = h(x)/q(x)$

Population density

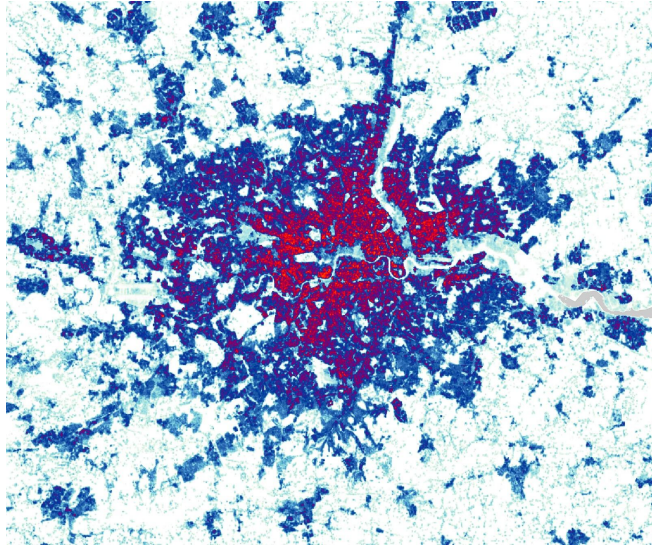
- What happens to population density as distance to the CBD increases ($x \uparrow$)?
- We have already seen that in the Alonso model, $x \uparrow D \downarrow$
- How about in the Muth-Mills model?
 - Intuitively we expect population density to decrease far from CBD
 - Formally (compute derivative of a ratio, i.e. using quotient rule):

$$\frac{\partial D(x)}{\partial x} = \frac{h'(x)q(x) - h(x)q'(x)}{[q(x)]^2} < 0$$

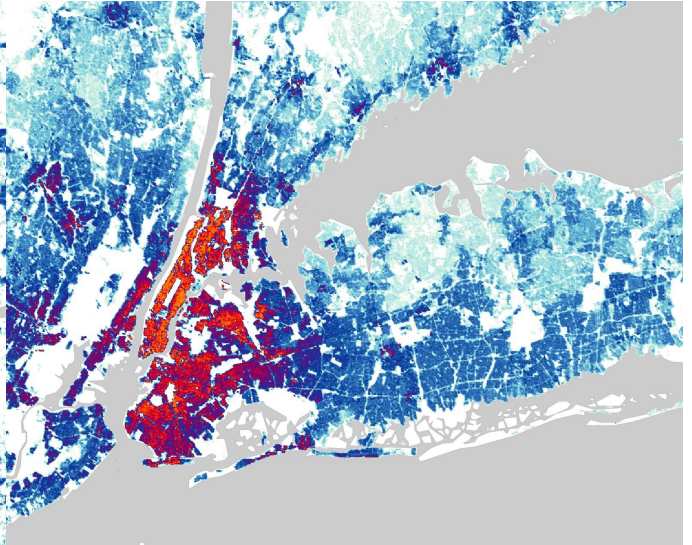
- Term in orange < 0 ; terms in purple > 0

Population density around the world

Population density in London



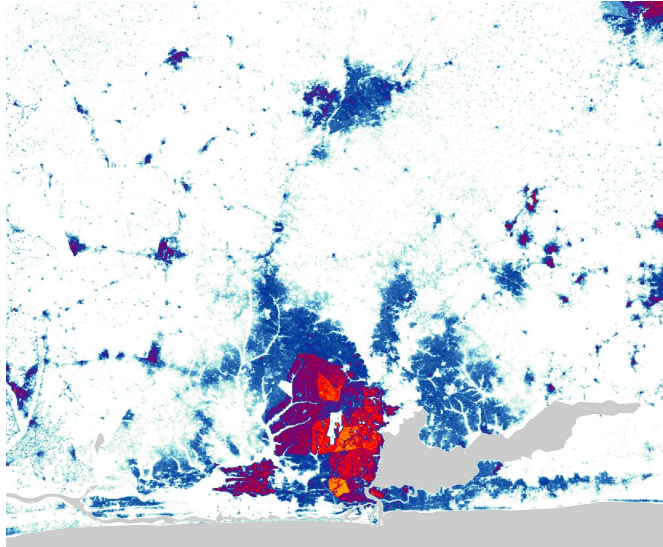
Population density in New York



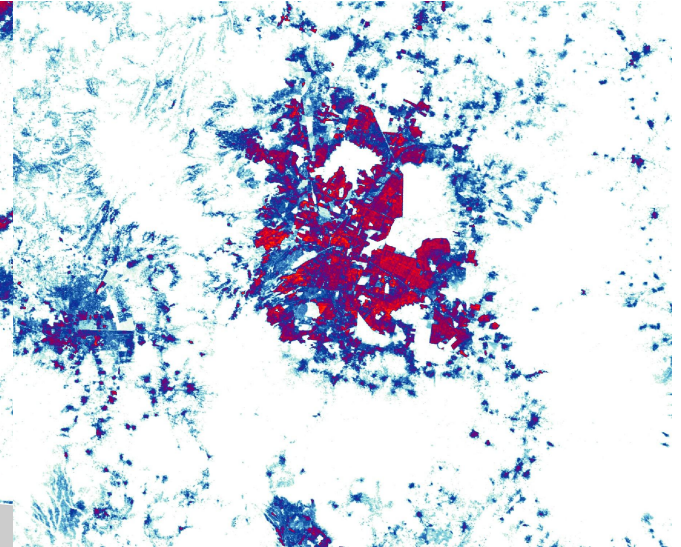
Source: <https://luminocity3d.org/WorldPopDen/#3/20.00/10.00>

Population density around the world

Population density in Lagos



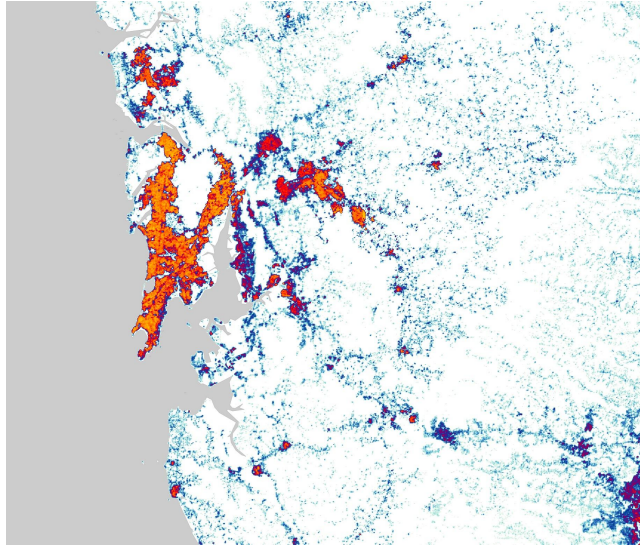
Population density in Mexico City



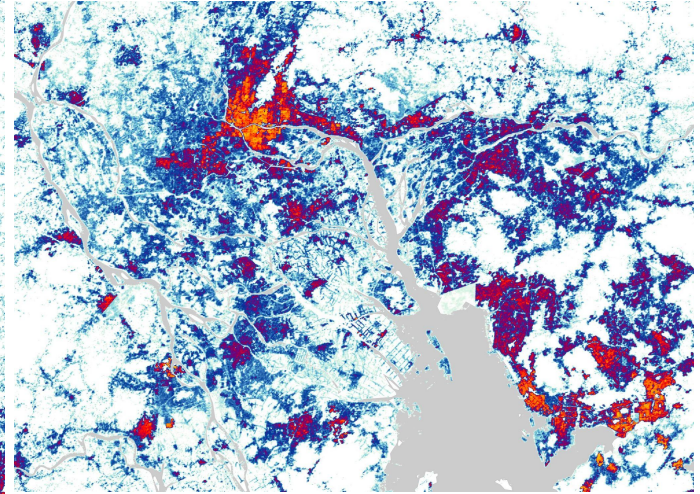
Source: <https://luminocity3d.org/WorldPopDen/#3/20.00/10.00>

Population density around the world

Population density in Mumbai

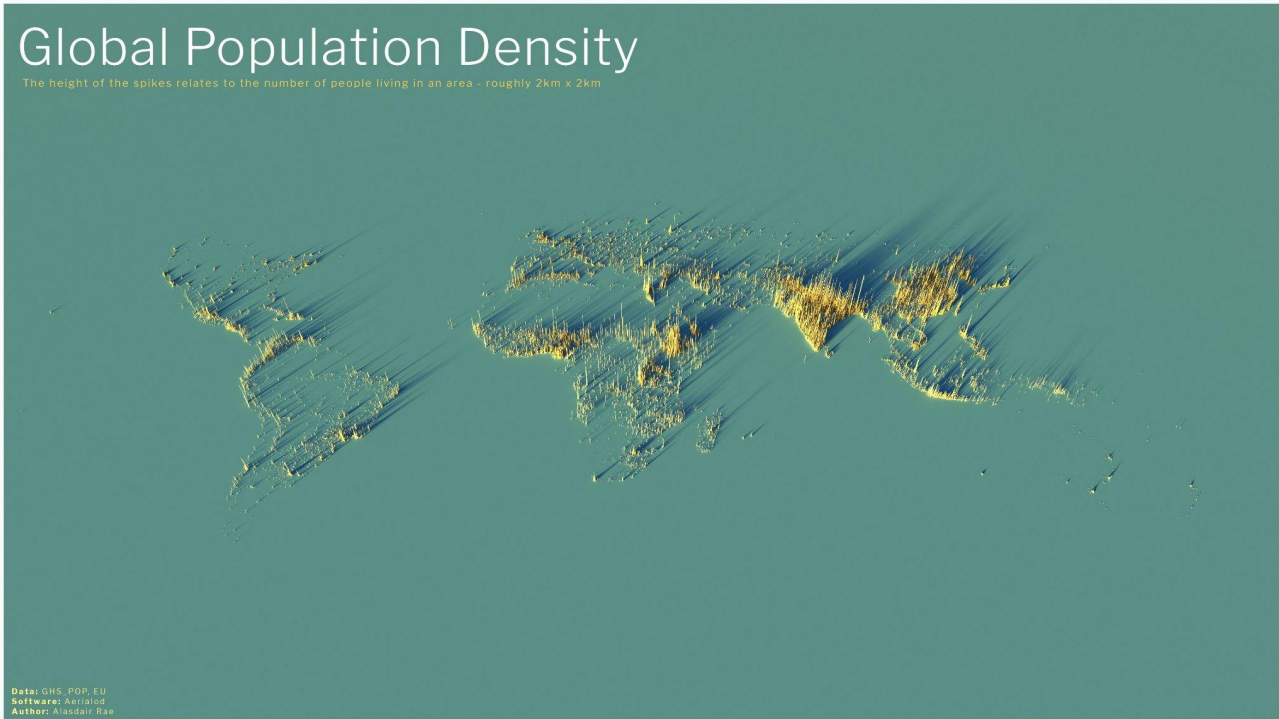


Population density in Guangzhou



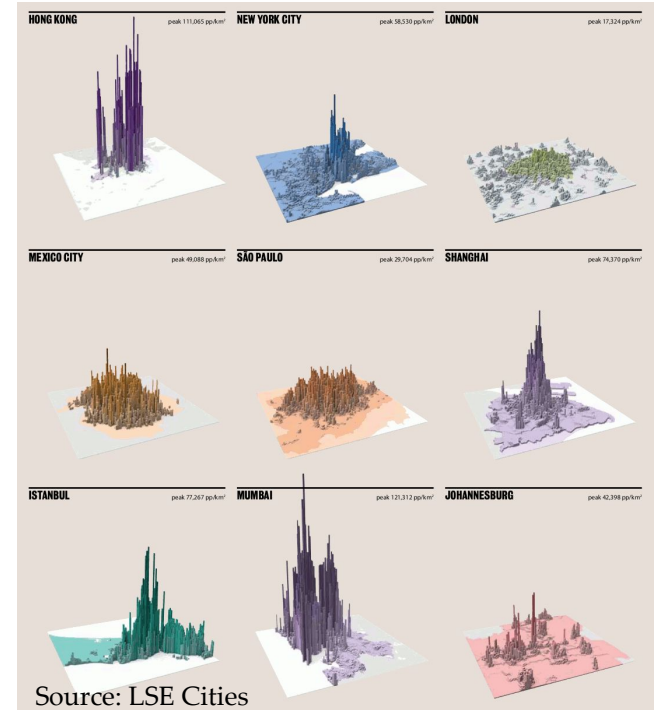
Source: <https://luminocity3d.org/WorldPopDen/#3/20.00/10.00>

Global population density



How relevant is the AMM model?

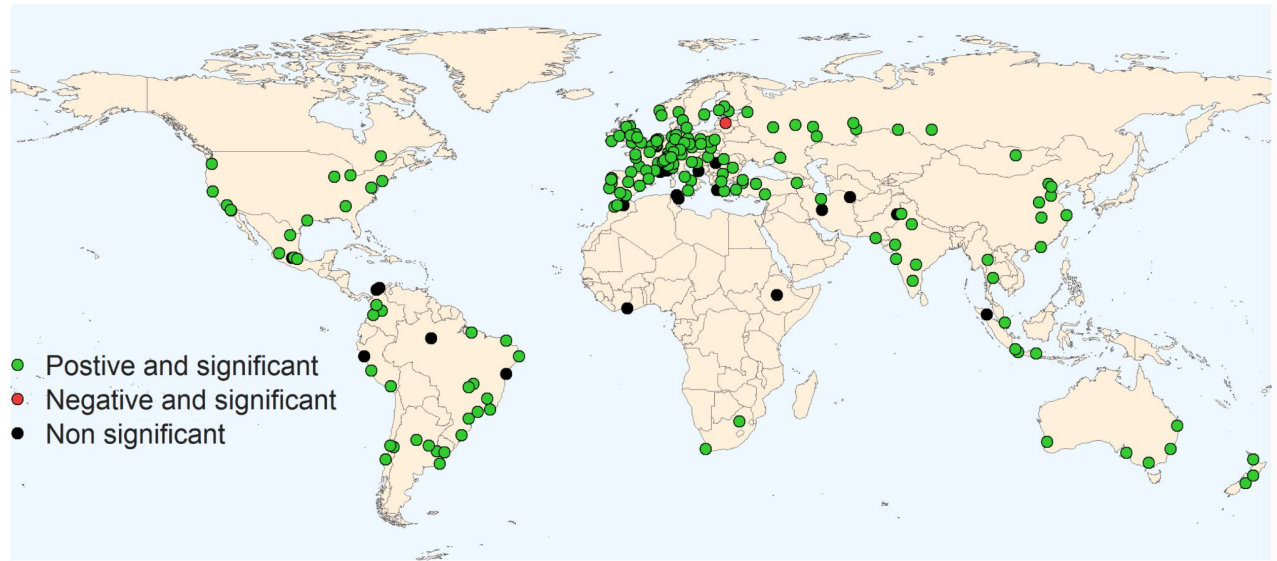
- Models are simplification of reality
 - A good model can be simple, but should be able to explain relevant features of the world around us
- How monocentric (centralized) are cities today?
- Which types of cities tend to be more centralized?



Liotta et al. (2022)

- Test the monocentric urban model in a global sample of cities
- Collect a large dataset of locations (1 sq km grid cells) within cities for 192 cities worldwide
 - Population density, rent, dwelling sizes, transportation times, land cover
- Estimate gradients for a global sample of cities
 - Population density gradient and rent gradient
- Analyze determinants of the heterogeneity in data

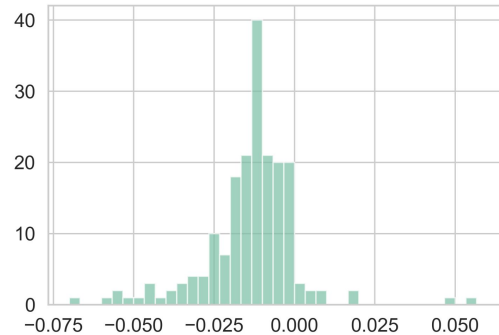
Liotta et al. (2022): Rent gradients



The Figure shows the sign of the relationship between accessibility and rent

Source: Liotta et al. (2022)

Liotta et al. (2022): Rent gradients



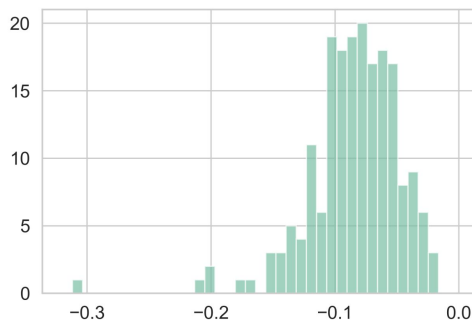
(a) Histogram - distribution of parameter f'_c .

	Mean	Min.	Q1	Med.	Q3	Max.	Nb. of obs.
OLS	-0.014	-0.070	-0.019	-0.012	-0.006	0.056	191

(b) Table - distribution of parameter f'_c .

- Most cities have rent gradient consistent with monocentric city
- On average, rent decreases by around 1.4% per km distance from the CBD
 - But there is heterogeneity

Liotta et al. (2022): Population density gradients



(a) Histogram - distribution of parameter h'_c .

	Mean	Min.	Q1	Med.	Q3	Max.	Nb. of obs.
OLS	-0.085	-0.309	-0.102	-0.081	-0.060	-0.017	192

(b) Table - distribution of parameter h'_c .

- On average, population density decreases by about 8.5% for every extra km distance from CBD
 - But heterogeneity
 - For instance, flatter in North America, Oceania, or when city has larger population

Summary

- Alonso-Mills-Muth (AMM) model is workhorse tool in urban economics
 - Assumes all employment is concentrated in the city center (CBD)
 - Spatial equilibrium implies lower rents at higher commuting distances
 - Housing rent decreases with distance to CBD
 - But: Housing consumption increases with distance to CBD
- Developers respond to changes in housing rent
 - Land rent decreases in distance to the CBD
- In real world, most cities tend to be centralized, but there is heterogeneity

Next lecture

- **THIS WEEK!** Wed 2-4 pm (PAR.LG.03)
- Monocentric city model II
 - Equilibrium conditions and additional restrictions
- Comparative statics: Income
 - City structure and income: Income segregation
- Comparative statics: Transportation costs
 - City structure and transportation costs: Decentralization
- Other predictions
 - Open city vs. closed city

Thank you!

Readings and references

Required readings

- DiPasquale and Wheaton, Ch. 3, The Urban Land Market: Rents and Prices.
- McDonald and McMillen, Ch. 6, Introduction to urban location patterns: Static analysis; Ch.7, Using the Monocentric City Model.

Other readings and references

- Ahlfeldt, G. M., and McMillen, D. P. (2018). Tall Buildings and Land Values: Height and Construction Cost Elasticities in Chicago, 1870 –2010. *Review of Economics and Statistics*, 100(5): 861-875.
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- Liotta, C. Viguie, V., Lepetit, Q. (2022), Testing the monocentric standard urban model in a global sample of cities. *Regional Science and Urban Economics*.
- Mills, E. (1967). An Aggregative Model of Resource Allocation in a Metropolitan Area. *American Economic Review*, 57(2), 197–210.

Readings and references

- Muth, R. (1969). Cities and Housing. Chicago: University of Chicago Press.
- Ricardo, D. (1817): On the Principles of Political Economy and Taxation. John Murray
- Von Thünen, J. (1826). Der isolierte Staat.